CRICKSTATS APPLICATION REPORT

Comprehensive Implementation & System Design Documentation

Executive Summary

The **CRICKSTATS** project is a dedicated Salesforce-based Cricket Player Performance Tracking Application designed to capture, organize, and analyze detailed, match-by-match player statistics. Developed for coaches, team managers, sports analysts, and administrators, the system addresses the inefficiencies of manual record-keeping, data fragmentation, and delayed analytics by delivering a unified cloud platform.

1. Project Overview & Objectives

CRICKSTATS digitizes cricket records, improves data accuracy, and facilitates real-time performance tracking. By leveraging custom Salesforce objects, master-detail relationships, roll-up summaries, and automated flows, the platform empowers stakeholders to make strategic selection decisions and execute data-driven player development initiatives.

Key Objectives

- **Digitize Player Records:** Centralize player profiles, stats, and match performance details in Salesforce.
- **Real-Time Analytics:** Deliver instant reporting and dashboard visualizations for coaches and analysts.
- **Automate Data Flows:** Eliminate manual updates and record inconsistencies using Salesforce Flow Builder.
- **Scalable Infrastructure:** Provide secure, cloud-hosted access capable of scaling across teams and leagues.

2. Ideation & Empathy Mapping

2.1 Problem Statements

- **PS-1 (Coach):** Struggles to track player performance accurately across matches due to data being scattered across different files and formats, leading to frustration and uncertainty in team selection.
- **PS-2 (Manager):** Experiences high stress and inefficiency while attempting to maintain complete player records manually due to excessive time spent updating and verifying data.
- **PS-3 (Sports Analyst):** Discovers comparing performances and evaluating player trends difficult because there is no centralized analytics system.

2.2 Empathy Map Canvas

- **Think & Feel:** Needs accurate, consolidated data and improved analytics tools.
- **Hear:** Management feedback requesting data-driven decision-making and streamlined data collection.
- **See:** Scattered spreadsheets, disconnected systems, and inconsistent record formats.
- **Say & Do:** Manually gathers, checks, and re-checks player statistics.
- **Pains:** Hours lost in verification, report errors, and data untrustworthiness.
- **Gains:** Automated workflows, quick decision-making, and trusted analytics dashboards.

2.3 Selected Solution

A custom **Salesforce-integrated CRM application** featuring custom objects, automated workflows, and Lightning dashboards.

3. Requirements Analysis & System Architecture

3.1 Functional Requirements (FRs)

- **FR-1 Player Registration:** Formally enroll and maintain player profile attributes.
- **FR-2 Match Performance Entry:** Capture detailed match statistics per player.
- **FR-3 Dashboard Generation:** Visualize core metrics like runs, wickets, and averages.
- **FR-4 Report Creation:** Filter and export consolidated player performance metrics.
- **FR-5 Flow Automation:** Automate UI interactions and player selection flows.

3.2 Key Non-Functional Requirements (NFRs)

- **Usability (NFR-1):** Intuitive Lightning UI requiring minimal user training.
- **Security (NFR-2):** Role-based access control via Profiles and Permission Sets.
- **Reliability (NFR-3):** Automatic data integrity checks and cloud backup protection.
- **Performance (NFR-4):** Dashboard and report page loads completing within 2–3 seconds.
- **Availability (NFR-5):** High 24/7 cloud uptime for continuous tournament access.
- **Scalability (NFR-6):** Infrastructure scales seamlessly from single-team to multi-league tracking.

3.3 Solution Architecture

The application uses a 4-tier Salesforce Architecture:

1. **Presentation Layer:** Salesforce Lightning Experience User Interface.
2. **Application Layer:** Salesforce Screen Flows and business logic rules.
3. **Data Layer:** Custom Objects (Player__c and Match_Performance__c).
4. **Analytics Layer:** Native Salesforce Reports & Interactive Dashboards.

4. Technical Data Model & Schema

4.1 Custom Objects & Fields

Object 1: Player__c (Parent Object)

Record Name: Player Name (Data Type: Text)
Optional Features: Allow Reports, Allow Activities, Track Field History, Allow Search

Field Label	Field API Name	Data Type	Details / Picklist Values
Player Name	Player_Name__c	Text	Length: 50

Field Label	Field API Name	Data Type	Details / Picklist Values
Player Number	Player_Number__c	Text	Length: 3
Profile Picture	Profile_Picture__c	Rich Text Area	Image storage
Date of Birth	Date_of_Birth__c	Date	Date picker
Role	Role__c	Picklist	Batsman, Bowler, Batsman & Wicket Keeper, All Rounder
Batting Style	Batting_Style__c	Picklist	Right-Handed, Left Handed
Bowling Style	Bowling_Style__c	Picklist	Right-Arm Fast, Left-Arm Fast, Right-Arm Medium, Left-Arm Medium, Right-Arm Spin, Left-Arm Spin
Nationality	Nationality__c	Picklist	Australia, India, England, South Africa, New Zealand, Pakistan, etc.
Team	Team__c	Picklist	CSK, DC, KKR, MI, PBKS, RR, RCB, SRH, GT, LSG
AVG Score	AVG_Score__c	Formula (Number)	IF(Matches_Played__c > 0, Total_Runs__c / Matches_Played__c, 0)

Object 2: Match_Performance__c (Child Object)

Record Name: Customer Account Name / Match Performance Name (Data Type: Text, Length: 80)
Optional Features: Allow Reports, Allow Activities, Track Field History, Allow Search

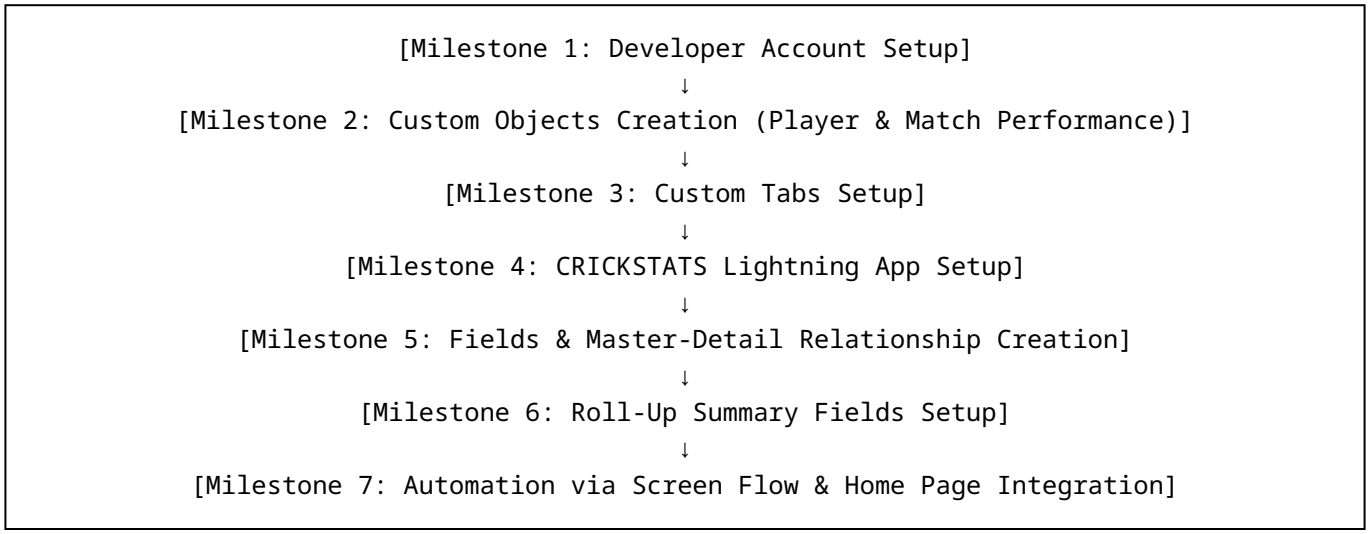
Field Label	Field API Name	Data Type	Details
Player	Player__c	Master-Detail	Related to Player__c
Player of the Match	Player_of_the_Match__c	Checkbox	True/False
Match Result	Match_Result__c	Picklist	WON, LOST, DRAW
Runs Scored	Runs_Scored__c	Number	Input score
Balls Faced	Balls_Faced__c	Number	Input count
Fours	Fours__c	Number	Input count
Sixes	Sixes__c	Number	Input count
Overs Bowled	Overs_Bowled__c	Number	Input count
Wickets Taken	Wickets_Taken__c	Number	Input count
Runs Conceded	Runs_Conceded__c	Number	Input count
Maiden Overs	Maiden_Overs__c	Number	Input count
Stumpings	Stumpings__c	Number	Input count
Catches	Catches__c	Number	Input count

4.2 Roll-Up Summary Fields on Player__c

Configured to summarize data from child Match_Performance__c records:

Field Label	Summarized Object	Roll-Up Type	Target Field / Filter Criteria
Matches Played	Match Performances	COUNT	All records included
Number of 50's	Match Performances	COUNT	Filter: Runs_Scored__c ≥ 50 AND Runs_Scored__c < 100
Number of 100's	Match Performances	COUNT	Filter: Runs_Scored__c ≥ 100 AND Runs_Scored__c < 200
Total 4's	Match Performances	SUM	Field to Aggregate: Fours__c
Total 6's	Match Performances	SUM	Field to Aggregate: Sixes__c
Total Runs	Match Performances	SUM	Field to Aggregate: Runs_Scored__c
Total Overs	Match Performances	SUM	Field to Aggregate: Overs_Bowled__c
Total Wickets	Match Performances	SUM	Field to Aggregate: Wickets_Taken__c
Total Stumpings	Match Performances	SUM	Field to Aggregate: Stumpings__c
Total Catches	Match Performances	SUM	Field to Aggregate: Catches__c

5. Implementation Milestones



Milestone Steps Summary

1. **Developer Org Setup:** Register at developer.salesforce.com/signup, verify email, set credentials, and open Setup.
2. **Object Creation:** Create Player and Match Performance objects with activity and reporting capabilities enabled.
3. **Tabs Setup:** Create Custom Object Tabs for navigation.
4. **Lightning App Configuration:** Build CRICKSTATS application via App Manager, selecting standard navigation along with Players, Match Performances, Reports, and Dashboards tabs.
5. **Data Architecture Definition:** Build custom fields, formula fields, and establish the Master-Detail relationship with Player as parent.

6. **Roll-up Summaries:** Add aggregation fields to Player object to track totals, fifties, and centuries automatically.
7. **Screen Flow Build (Player_Flow):**
   - **Screen Step 1:** Picklist for Team selection.
   - **Get Records:** Filter Player__c records matching selected Team.
   - **Screen Step 2:** Display results in a Data Table featuring columns: Profile Picture, Player Name, Total Runs, AVG Score, Batting/Bowling Styles, Matches Played, 50s, and 100s.
   - **Deployment:** Embed the flow directly into the CRICKSTATS Home Page layout.

6. Planning, Testing & Evaluation

6.1 Sprint Schedule & Project Tracking

Sprint	Functional Epic	User Story	Story Points	Priority
Sprint-1	Developer Setup / Data Modeling	Setup Dev Org (USN-1), Create Objects (USN-2), Object Tabs (USN-3)	7	High
Sprint-2	UI Design / Data Structure	Dedicated Lightning App (USN-4), Setup Fields & Relationships (USN-5), Page Layouts (USN-6)	13	High/ Medium
Sprint-3	Logic & Automation	Validation Rules (USN-7), Player Selection Flow (USN-8)	8	High
Sprint-4	Analytics & Deployment	Reports, Dashboards & User Acceptance Testing	12	High

6.2 Testing & Validation Strategy

- **Functional Validation:** Verified record creation, master-detail cascade behaviors, and validation rules.
- **Automation Testing:** Executed Player_Flow scenarios across multiple teams to ensure table dynamic filtering functions correctly.
- **Roll-up Accuracy:** Calculated manual test scenarios against automatically calculated aggregate values.
- **Performance Testing:** Evaluated UI responsiveness, confirming dashboards render within target thresholds.

7. Trade-off Analysis & Future Roadmap

7.1 Advantages & Disadvantages

Advantages

- **Centralized Database:** Replaces scattered spreadsheets with a single source of truth.
- **Real-time Performance Metrics:** Automated Roll-up Summary fields provide live statistics instantly.
- **Enterprise Security:** Utilizes Salesforce's robust security architecture and access permissions.
- **Scalable Infrastructure:** Broadens easily from individual teams to global tournament management.

Disadvantages

- **Licensing Requirements:** Dependent on active Salesforce licenses.
- **Internet Dependency:** Requires stable internet connectivity for cloud access.
- **Limited Native Offline Access:** Dynamic web flows require real-time connection.

7.2 Strategic Future Roadmap

1. **Mobile Application Integration:** Tailor views for mobile devices using Salesforce Mobile layout optimizations.
2. **AI Predictive Analytics:** Integrate Einstein Analytics to predict player form trends and performance probabilities.
3. **Video Analysis Integration:** Link video clips directly to specific match performance records.
4. **Public Statistics Portal:** Expose non-sensitive player statistics to fans using Salesforce Experience Cloud.